

Achref Lajmi

Software Engineer — Backend, Full-Stack & Applied AI

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PROFILE

Software engineering graduate with production internship experience designing and shipping AI-driven systems end to end. As the sole developer of the AI layer during a six-month internship at a Munich B2B marketplace startup, I built an autonomous deal engine and a hybrid semantic search engine now running in production. Comfortable owning a domain from data model to background jobs to user-facing dashboards. Seeking **Backend, Full-Stack or Applied AI Engineer** roles.

Work eligibility: Already resident in Germany — no relocation required. Degree is Master-equivalent (EQF level 7), meeting the EU Blue Card qualification requirement; employer support needed to convert my current residence permit to a work permit.

EXPERIENCE

Software Engineering Intern (End-of-Studies Project) – recash / Circular Service GmbH Feb 2026 – Jul 2026
Munich, Germany | Next.js, TypeScript, PostgreSQL/Supabase, Drizzle ORM, pgvector, Trigger.dev, Claude API via OpenRouter, Vercel

Sole developer of the platform's AI layer in a nine-person startup. Owned design and delivery across 24 sprints under continuous deployment, with CTO code review.

- Designed and built **SmartPilot**, an autonomous deal engine closing a four-stage loop: nightly inventory health scoring, AI-mediated price negotiation, targeted deal distribution, and a feedback-driven learning loop. Scored 200 products nightly and queued 61 pricing opportunities with no manual initiation.
- Replaced keyword search with a **three-signal hybrid retrieval engine** (pgvector embeddings, PostgreSQL full-text, keyword ranking) — shipped to production, returning relevant results for queries sharing no literal words with the listing.
- Built a behavioural recommendation engine deriving a per-buyer taste vector from interaction history, sustaining a 25–35% discovery fraction under an enforced brand-diversity floor; later inverted it to match live deals to the buyers most likely to want them.
- Implemented a nightly **learning loop** adjusting matching weights from buyer feedback within fixed safety bounds — autonomously moved the size-fit weight from 0.10 to 0.42, never overwriting a human-set value.
- Shipped an agentic AI assistant (Claude via Vercel AI SDK) grounded in live marketplace data through a fixed, user-scoped tool menu preventing the model from reaching data outside the caller's scope.
- Delivered a production eBay connector (REST and Trading APIs) plus code-complete Shopify Admin and Amazon SP-API integrations; rebuilt order management to handle all four financial edit scenarios correctly.
- Enforced human-in-the-loop guarantees and auditability: no price reaches the catalogue without an administrator decision stamped on the record it concerns, backed by append-only ledgers and row-level access scoping.

AI Software Developer Intern – MASMEDIA (ESPRIT Tech) Jul 2025 – Sep 2025
Tunis, Tunisia | NestJS, TypeScript, MongoDB, GPT-4, AssemblyAI, FFmpeg, Streamlink, Cheerio, Jest, Docker
Sole developer of MASMEDIA AIDITOR, an AI journalism platform built for a tech newsroom, delivered across two modules in Agile sprints.

- Built **PARROT**, a real-time livestream coverage pipeline (Streamlink audio extraction, AssemblyAI streaming transcription, GPT-4 key-moment detection, multi-platform content generation) reaching **3.2s transcription latency** and **84% detection precision**.
- Validated the pipeline live on real events: 8 of 8 announcements detected on an Apple product keynote with a **73% cut in coverage time**, and 15 of 16 on a Google I/O keynote.
- Built **SWALLO**, a configurable scraper (Axios + Cheerio) over 30 technology sources with GPT-4 relevance filtering — processed 1,247 articles across a seven-day run at 84% precision.
- Reduced event coverage from **4h30 to 1h20** and daily news monitoring from **4 hours to 12 minutes**, cutting the team needed per live event from three people to one.
- Automated publishing to Facebook, Instagram and Twitter over OAuth 2.0 with encrypted token storage and pre-expiry refresh; added exponential-backoff retries and a response cache that cut GPT-4 API spend by 35%.
- Streamed live transcripts and detected moments to the UI over Server-Sent Events; shipped 95 unit and integration tests at 88% coverage (Jest, Supertest, in-memory MongoDB).

Full-Stack Developer Intern – FinConnect Financial Solutions Jun 2024 – Jul 2024
Tunis, Tunisia | Angular, Spring Boot, PostgreSQL, JWT
Delivered features on a secure internal ticketing platform used by the company's support teams.

- Implemented JWT authentication and role-based access control separating agent, manager and administrator permissions across the application.
- Built ticket filtering, reporting and PDF export workflows, plus an in-app chat channel between requesters and support agents.
- Worked in Scrum sprints with peer code review, shipping features against fixed deadlines.

Tunis, Tunisia | Spring Boot, Angular

First professional development role, on a Scrum-based delivery team.

- Built backend REST endpoints and responsive Angular UI components for client-facing modules.
- Refactored existing modules to remove duplication and improve maintainability.

EDUCATION

ESPRIT – Private Higher School of Engineering and Technology, Tunis, Tunisia

Sep 2021 – Sep 2026

Diplôme d'Ingénieur in Computer Science – Web & Mobile Application Development – **graduated with highest honours (mention Très Bien)**

Five-year integrated engineering programme. EQF level 7 — equivalent to a German Master of Science.

Philipps-Universität Marburg, Germany

Oct 2025 – Mar 2026

Exchange Semester – Computer Science

TECHNICAL SKILLS

Languages: TypeScript / JavaScript, Java, Python, SQL

Backend: Node.js, Next.js server actions & API routes, NestJS, Spring Boot, REST APIs, Zod, JWT / RBAC

Frontend: React, Next.js, Angular, Tailwind CSS

Data: PostgreSQL, pgvector, Supabase, Drizzle ORM, Prisma, MongoDB, Row-Level Security, data modelling

Applied AI: LLM integration, RAG, tool-calling agents, embeddings, semantic & hybrid search, recommender systems

Infrastructure: Docker, Git/GitHub, CI/CD, Turborepo + pnpm monorepos, Trigger.dev, Vercel, Sentry, PostHog

SELECTED PROJECTS

Kiddo AI – AI Learning Companion *Flutter, Spring Boot, MongoDB, OpenAI, STT/TTS | five-person academic team*

- Educational app for Tunisian first-year primary pupils (age 6), with an AI tutor speaking **Tunisian Arabic**, cloned-voice synthesis and avatar-based personalisation.
- **Principal author of the client's second iteration:** restructured the prototype into MVVM, extracted the service and model layers, and rewrote the home, lesson and chat screens — roughly 3,100 of the client's 4,400 lines.
- Integrated the voice loop across three runtimes (Flutter client, Spring Boot API, external inference service), including the decode chain that keeps Arabic text intact through transcription, generation and speech synthesis.

MoodyMap – Mood-Aware Study Planner (iOS)

SwiftUI, MVVM, async/await, Swift Charts, URLSession

- Built the emotion-capture pipeline: hand-assembled RFC 2388 multipart uploads over `URLSession` with JPEG re-compression and a five-branch error taxonomy, feeding a server-side facial-emotion model.
- Bridged camera and photo-library permissions across AVFoundation and PhotosUI — two permission systems with different state vocabularies — into SwiftUI through a `UIViewControllerRepresentable` coordinator.
- Parallelised the home screen with `withTaskGroup`, loading profile, mood statistics, recommendations, notifications and a daily quote concurrently with independent failure handling.
- Turned unstructured LLM study-plan prose into an interactive, checkable task list — consuming non-deterministic model output with no schema. 10 screens, 18 endpoints, 5,000 lines of Swift.

SUF'ESS – Cultural Exploration App

FlutterFlow, RAG, 3D, Geolocation | 24-hour hackathon

- Shipped a working location-based storytelling MVP with prototype AI guided-tour features in a single 24-hour build.
- Scoped the build to a demonstrable core under a hard deadline, trading breadth for a path that ran end to end on stage.

LANGUAGES

Arabic: Native French: C1 English: C1 (TOEFL iBT) German: A1, actively studying

CERTIFICATIONS

AWS Academy Cloud Foundations | NVIDIA Generative AI with Diffusion Models | NVIDIA Applications of AI for Predictive Maintenance